MINIMUM LAP TIME OPTIMIZATION

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Abstract.

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1. The problem setup

he objective of this project is to compute the minimum-time trajectory for a race car on a closed racing circuit. Unlike conventional path-following problems, where the trajectory is predefined, both the vehicle trajectory and the control inputs are optimized simultaneously while satisfying the physical limitations of the vehicle and the geometric constraints imposed by the track.

The optimization is performed on a real karting circuit located in Freiburg. The track geometry is represented by its centerline, local curvature and track width, which constitute the geometric input to the optimal control problem. These quantities are obtained by processing an image of the circuit and extracting its boundaries.

Besides the track geometry, the relevant vehicle parameters are assumed to be known. These include the wheelbase, steering angle limits, longitudinal acceleration and braking limits, and the maximum admissible lateral acceleration. Together, these parameters define the dynamic and physical constraints of the optimization problem.

The objective is therefore to determine the state and control trajectories that minimize the total lap time while ensuring that the vehicle remains within the track boundaries and satisfies all vehicle constraints.

To formulate this optimal control problem, an appropriate mathematical model of the vehicle is required. In the following section, a simplified kinematic bicycle model is introduced together with the assumptions adopted throughout this work.

2. Vehicle model

In order to formulate our optimization problem, we first need to decide on some simplifications. The goal of this project is not to focus on the vehicular dynamics of a race car, but rather on the optimization problem itself.

Therefore, we decided to use the commonly used kinematic bicycle model to describe the dynamics of the car.

2.1. The bicycle model.

The commonly used kinematic bicycle model is a simplification of the dynamics of a car. It assumes that the car has two wheels, one in the front and one in the back. Only the front wheel is used for steering and the slip angle is neglected.

The model is described by the position of the car in global coordinates (x, y), the velocity v, the heading ψ and the steering angle δ. The dynamics of the model are given as

$$\begin{aligned} \dot{x} &= v\cos(\psi) \\ \dot{y} &= v\sin(\psi) \\ \dot{\psi} &= \frac{v}{L}\tan(\delta) \\ \dot{v} &= a. \end{aligned}$$

The constants L and a denote the wheelbase of the car, i.e. the distance between the front and rear wheel, and the acceleration of the car respectively. For more details on this model, see [Raj12].

However, for driving around a given track it is inconvenient to use global coordinates because it is more complicate to formulate the constraints. Additionally, we only need the position on the track and do not care for the absolute position in global coordinates.

Therefore, we want to formulate the model using track coordinates. The centerline distance traveled will be given as s and e_y denotes the distance from the centerline, e_ψ denotes the angle between the cars heading and the tangent of the centerline and v still denotes the velocity of the vehicle.

Now, we can split the velocity into the forward component along the track tangent given as $v\cos(e_\psi)$ and the sideways component normal to the track tangent given as $v\sin(e_\psi)$, which directly leads to $\dot{e_y} = v\sin(e_\psi)$ because the lateral offset changes exactly by the sideways component of the velocity.

On a curved track, we need to take into account that a car with lateral offset to the centerline needs to travel a different distance compared to a car being on the centerline in order to achieve the same progress. This leads to the relation EXPLAIN!

$$\dot{s} = \frac{v\cos(e_\psi)}{(1 - \kappa(s)e_y)},$$

with $\kappa(s)$ being the curvature of the track at position s.

Now, for the heading deviation, we calculate

$$e_\psi = \psi - \psi_c(s)$$

and differentiating yields

$$\dot{e}_\psi = \dot{\psi} - \dot{\psi}_c(s)\dot{s}.$$

The cars own heading with respect to the tangent of the centerline changes because of steering and the heading of the track changes because of the curvature of the track. Therefore, we get

$$\dot{e}_\psi = \frac{v}{L}\tan(\delta) - \kappa(s)\dot{s}.$$

This yields the following model

$$z(t) = \begin{bmatrix} s(t) \\ e_y(t) \\ e_\psi(t) \\ v(t) \end{bmatrix} \qquad \dot{z}(t) = \begin{bmatrix} \dot{s}(t) \\ v(t)\sin(e_\psi(t)) \\ \frac{v}{L}\tan(\delta(t)) - \kappa(s(t))\dot{s}(t) \\ a(t) \end{bmatrix}$$

EXPLAIN LIMITATIONS OF THE MODEL

3. Optimization Problem

3.1. **Formulation of the optimization problem.**

We need to expand the section explaining carefully:

(1) Decision variables
(2) Objective
(3) Constraints
(4) Boundary conditions

In order to formulate the optimization problem, we need to have some controls of the car. These are given as

$$u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix}$$

with a being the acceleration and δ being the steering angle. With this, we can finally formulate the optimization problem. Using the model above, we can formulate the problem as follows

$$\underset{z(\cdot),\, u(\cdot),\, T}{\text{minimize}} \quad T \tag{3.1}$$

$$\text{subject to} \quad \dot{z}(t) = \begin{bmatrix} \dot{s}(t) \\ v(t)\sin(e_\psi(t)) \\ \frac{v}{L}\tan(\delta(t)) - \kappa(s(t))\dot{s}(t) \\ a(t) \end{bmatrix} =: f\big(z(t), u(t)\big), \qquad t \in [0, T], \tag{3.2}$$

$$-\frac{w\big(s(t)\big) - b}{2} \le e_y(t) \le \frac{w\big(s(t)\big) - b}{2}, \tag{3.3}$$

$$-a_{\min} \le a(t) \le a_{\max}, \tag{3.4}$$

$$-\delta_{\max} \le \delta(t) \le \delta_{\max}, \tag{3.5}$$

$$0 \le v(t) \le v_{\max}, \tag{3.6}$$

$$\left| \frac{v(t)^2}{L} \tan\big(\delta(t)\big) \right| \le a_{y,\max}, \tag{3.7}$$

$$s(0) = 0, \qquad s(T) = S, \tag{3.8}$$

$$v(0) = 0, \qquad e_y(0) = 0, \qquad e_\psi(0) = 0. \tag{3.9}$$

The constants used in the definition of the optimization problem are defined as follows:

$w(s)$: track width at the longitudinal position s,
b: width of the vehicle,
L: wheelbase of the vehicle,
$a_{\min}$: magnitude of the maximum permissible deceleration,
$a_{\max}$: maximum permissible acceleration,
$\delta_{\max}$: maximum absolute steering angle,
$v_{\max}$: maximum permissible velocity,
$a_{y,\max}$: maximum permissible lateral acceleration,

S: total length of the track.

4. Numerical solution

We need to cover here:

(1) Discretization
(2) Multiple-shooting
(3) CasAdi
(4) IPOPT (or other solver use)

References

[Raj12] Rajesh Rajamani. *Vehicle Dynamics and Control*. 2nd ed. Springer US, 2012. ISBN: 9781461414339. DOI: 10.1007/978-1-4614-1433-9.